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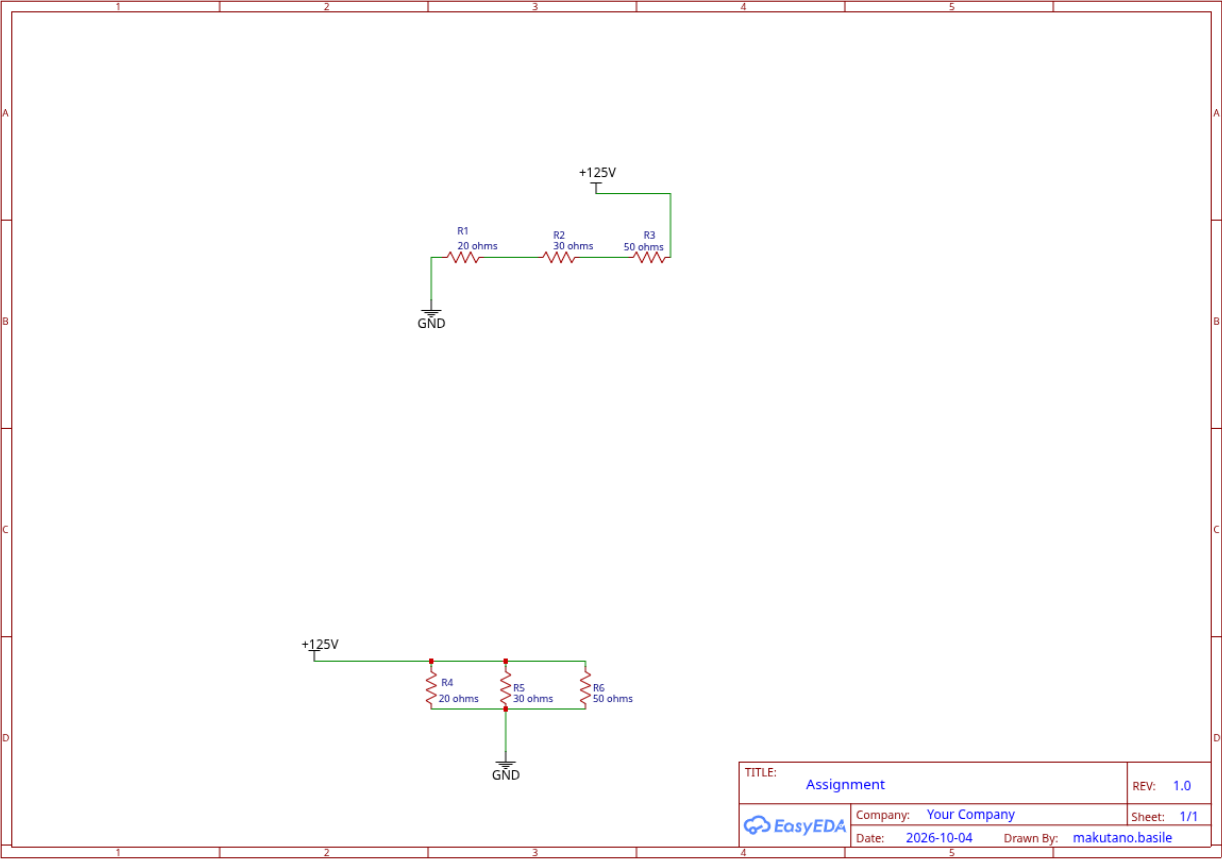
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DATE

M T W T F S S

PART A.

$$1. R_{tot} = R_1 + R_2 + R_3$$

$$= 20 + 30 + 50$$

$$= 100 \Omega$$

$$2. I_{tot} = \frac{V}{R_{tot}} = \frac{125 V}{100} = 1,25 A$$

3. $I_1 = I_2 = I_3 = I_{tot}$ since they are in series

$$4. V_1 = I_{tot} \times R_1 = 1,25 \times 20 = 25 V$$

$$V_2 = I_{tot} \times R_2 = 1,25 \times 30 = 37,5 V$$

$$V_3 = I_{tot} \times R_3 = 1,25 \times 50 = 62,5 V$$

$$5. P_1 = V_1 \times I_{tot} = 31,25 W$$

$$P_2 = V_2 \times I_{tot} = 46,875 W$$

$$P_3 = V_3 \times I_{tot} = 78,125 W$$

TAKE CIPLADON.THE FIZZ WORKS TO RELIEVE PAIN FFFASTER¹
¹Eur J Clin Pharmacol. 2000 May; 56(2): 141-3doi: 10. 1007 / s002280050732.
IF PAIN PERSIST ALWAYS CONSULT YOUR DOCTOR.

OF TRUSTED PARTNERSHIP
IN PAIN RELIEF IN KENYA
AND BEYOND!
Cipladon

1. Calculate the total resistance in Ω

$$\frac{1}{R_{\text{tot}}} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50}$$

$$0.05 + 0.01 + 0.02$$

$$= 0.08$$

$$R_{\text{tot}} = \frac{1}{0.08}$$

$$= 12.5 \Omega$$

2. Calculate total current from power supply

$$I_{\text{tot}} = \frac{V}{R_{\text{tot}}} = \frac{12.5}{12.5}$$

$$= 1 \text{ A}$$

3. Calculate current through each resistor

$$6.25 + 1.25 + 2.5 = 10$$

$$I_{\text{total}} = \frac{V}{R}$$

$$I_1 = \frac{12.5}{20} = 6.25 \text{ A}$$

$$I_2 = \frac{12.5}{100} = 1.25 \text{ A}$$

$$I_3 = \frac{12.5}{50} = 2.5 \text{ A}$$

4. Voltage drop across each resistor

It is similar as

$$V_1 = V_2 = V_3$$

$$= 12.5 \text{ V}$$

5. Calculate the power dissipated through each resistor (use Watt's law)

$$P = V \times C (I)$$

$$781.25 + 156.25 + 312.5 = 1250 \text{ W}$$

$$12.5 \times 6.25 = 781.25 \text{ W}$$

$$12.5 \times 1.25 = 156.25 \text{ W}$$

$$12.5 \times 2.5 = 312.5 \text{ W}$$